

V Measurement protocol

Comparison of methods for determining the period of oscillation

No.	number of oscillations n	measured time t [s]	period of oscillation T [s]	mean value $\bar{T}$ [s]	$\sigma_{\bar{T}}$ [s]
1	3	4,65	1,54	1,587	2,2 · 10 ⁻³
2	3	4,69	1,56		
3	3	4,81	1,60		
4	3	4,85	1,61		
5	3	4,76	1,59		
6	3	4,77	1,59		
7	3	4,77	1,59		
8	3	4,88	1,63		
9	3	4,70	1,57		
10	3	4,76	1,59		

Start and stop at the maximum deflection

No.	number ob oscillations n	measured time t [s]	period of oscillation T [s]	mean value $\bar{T}$ [s]	$\sigma_{\bar{T}}$ [s]
1	3	5,02	1,67	1,67	1,8 · 10 ⁻³
2	3	5,04	1,68		
3	3	5,04	1,68		
4	3	5,00	1,67		
5	3	5,01	1,67		
6	3	5,02	1,67		
7	3	5,02	1,67		
8	3	4,98	1,66		
9	3	5,00	1,67		
10	3	5,00	1,67		

Start and stop at the zero crossing

Accuracy of reading the stop watch: $\frac{1}{100}$ s

Measuring the spring constant

m [g]	Nr.	number of oscillations n	measured time t [s]	period of oscillation T [s]	mean value $\bar{T}$ [s]	$\sigma_{\bar{T}}$ [s]
50	1	3	2,70	0,90	0,902	2,5 · 10 ⁻²
	2	3	2,84	0,95		
	3	3	2,62	0,87		
100	1	3	3,76	1,25	1,255	9 · 10 ⁻³
	2	3	3,63	1,22		
	3	3	3,70	1,23		
150	1	3	4,48	1,49	1,498	5 · 10 ⁻³
	2	3	4,50	1,50		
	3	3	4,50	1,50		
200	1	3	5,04	1,68	1,674	5 · 10 ⁻³
	2	3	5,02	1,67		
	3	3	5,01	1,67		
250	1	3	5,55	1,84	1,844	1,8 · 10 ⁻²
	2	3	5,46	1,82		
	3	3	5,65	1,88		

Measuring the period of oscillation as a function of the mass.
Time measurement started and stopped at ...

Measurement of the gravitational acceleration

Δm [g]	deflection x [mm]	error of reading Δx [mm]
0	177	1
50	342	1
100	505	1
150	669	1
200	830	1
250	985	1

Measurement of deflection as a function of mass